Tech Support

Software version:	2.5.1K5
User name:	UTESE
Time and Date:	2025-03-10 20:00:32
Tray number:	14
Tray name:	Redo_COMPASS_Box7_MonMon_PO4.NOx.SO4_385-465_Feb202

Sulfate

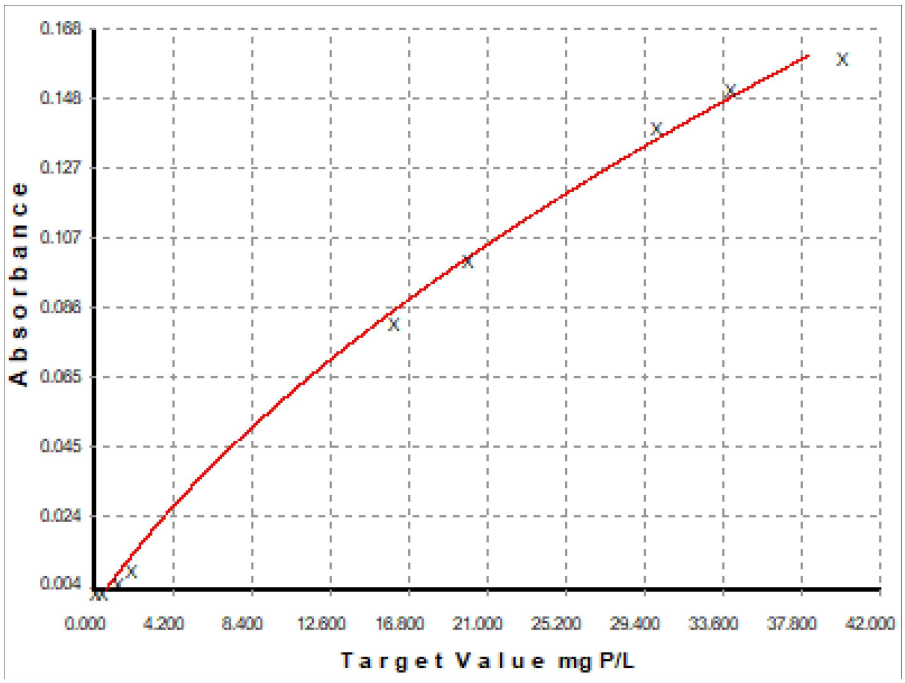
Calibration data

ID	Absorbance	Calc mg SO4/L	Target mg SO4/L	% Error
S1	0.0036	0.61	0.00	
S90	0.0037	0.62	0.40	55.3486
S91	0.0063	0.96	1.20	-19.8323
S92	0.0102	1.51	2.00	-24.6129
*93	0.0282	4.24	6.00	-29.3263
*94	0.0507	8.32	10.00	-16.8098
S95	0.0828	15.33	16.00	-4.1625
S96	0.1009	19.91	20.00	-0.4431
S97	0.1398	31.31	30.00	4.3761
S98	0.1516	35.17	34.00	3.4315
S99	0.1603	38.17	40.00	-4.5645
S0	0.0035	0.60	0.00	

Calibration status

Polynomial order	2
Correlation coefficient	0.9982
RSE (%)	28.75
Carryover(%)	-0.0
Calibration equation	y = cx^2 + bx + a
y =	Concentration mg SO4/L
x =	Measured absorbance
a =	1.482069E-01
b =	1.258099E+02
c =	6.945033E+02
Original Date & Time	2025-03-10 17:02:52 (2025-03-10 15:24:21)

Calibration graph



Tech Support

Reagents

Number	Reagent name	Batch number	User	Expiry date
4	Turbidimetric reagent (sulfate)		seal	
39	DI Water (Blank)		UTESE	

Test results

Cup	Type	ID	Details	Result	Units	Raw data	QC Pro	Auto dil.	Man dil.	User	T
	CCV	CCV		17.59	mg SO4/L	0.091964				UTESE	20
	CCB	CCB		0.57	mg SO4/L	0.003307				UTESE	20
11	U1	385	7_1_385_44967_OWC_OWC_W_PW_	41.47 10	mg SO4/L	0.050608		x5.0000		UTESE	20
21	U11	395	7_12_395_44967_OWC_OWC_TR_PW	64.82 A_20	mg SO4/L	0.143215		x2.0000		UTESE	20
22	U12	396	7_13_396_44967_OWC_OWC_TR_PW	52.58 A_45	mg SO4/L	0.061521		x5.0000		UTESE	20
22	U12	396		0.00	mg SO4/L	0.000000				UTESE	
25	U15	399	7_16_399_44967_OWC_OWC_TR_PW	46.65 B_45	mg SO4/L	0.055793		x5.0000		UTESE	20
25	U15	399		0.00	mg SO4/L	0.000000				UTESE	
28	U18	402 Sample 403 Empt	7_19_402_44967_OWC_OWC_TR_PW	57.98 C_45	mg SO4/L	0.066544		x5.0000		UTESE	20
32	U22	407	7_27_407_44967_OWC_OWC_UP_PW	51.52 C_10	mg SO4/L	0.121746		x2.0000		UTESE	20
45	U35	421	7_44_421_44967_OWC_OWC_UP_RZ	34.51	mg SO4/L	0.090640		x2.0000		UTESE	20
46	U36	422	7_45_422_44967_OWC_OWC_UP_RZ	46.86	mg SO4/L	0.113700		x2.0000		UTESE	20
48	U38	424	7_48_424_44967_OWC_OWC_UP_RZ	42.88	mg SO4/L	0.106559		x2.0000		UTESE	20
50	U40	426	7_51_426_44978_CRC_CRC_W_PW_	248.39 0	mg SO4/L	0.070275		x20.0000		UTESE	20
50	U40	426		0.00	mg SO4/L	0.000000				UTESE	
	CCV	CCV		17.92	mg SO4/L	0.093242				UTESE	20
	CCB	CCB		0.60	mg SO4/L	0.003544				UTESE	20
51	U41	427	7_52_427_44978_CRC_CRC_W_PW_	381.85 0	mg SO4/L	0.097789		x20.0000		UTESE	20
51	U41	427		0.00	mg SO4/L	0.000000				UTESE	
52	U42	428	7_53_428_44978_CRC_CRC_W_PW_	408.48 5	mg SO4/L	0.102812		x20.0000		UTESE	20
52	U42	428		0.00	mg SO4/L	0.000000				UTESE	
53	U43	429	7_54_429_44978_CRC_CRC_W_PW_	251.40 0	mg SO4/L	0.070948		x20.0000		UTESE	20
53	U43	429		0.00	mg SO4/L	0.000000				UTESE	
54	U44	430	7_55_430_44978_CRC_CRC_W_PW_	323.20 0	mg SO4/L	0.086226		x20.0000		UTESE	20
54	U44	430		0.00	mg SO4/L	0.000000				UTESE	
55	U45	431	7_56_431_44978_CRC_CRC_W_PW_	366.99 5	mg SO4/L	0.094929		x20.0000		UTESE	20
55	U45	431		0.00	mg SO4/L	0.000000				UTESE	
56	U46	432	7_57_432_44978_CRC_CRC_W_PW_	294.68 0	mg SO4/L	0.080322		x20.0000		UTESE	20
56	U46	432		0.00	mg SO4/L	0.000000				UTESE	
57	U47	433	7_58_433_44978_CRC_CRC_W_PW_	307.37 0	mg SO4/L	0.082973		x20.0000		UTESE	20
57	U47	433		0.00	mg SO4/L	0.000000				UTESE	
58	U48	434	7_59_434_44978_CRC_CRC_W_PW_	321.97 5	mg SO4/L	0.085975		x20.0000		UTESE	20
58	U48	434		0.00	mg SO4/L	0.000000				UTESE	
59	U49	435	7_61_435_44978_CRC_CRC_TR_PW_	171.04 10	mg SO4/L	0.148686		x5.0000		UTESE	20
60	U50	436	7_62_436_44978_CRC_CRC_TR_PW_	200.09 20	mg SO4/L	0.059077		x20.0000		UTESE	20
60	U50	436		0.00	mg SO4/L	0.000000				UTESE	
	CCV	CCV		17.79	mg SO4/L	0.092734				UTESE	20
	CCB	CCB		0.56	mg SO4/L	0.003241				UTESE	20
61	U51	437	7_63_437_44978_CRC_CRC_TR_PW_	236.11 45	mg SO4/L	0.067503		x20.0000		UTESE	20
61	U51	437		0.00	mg SO4/L	0.000000				UTESE	
62	U52	438	7_64_438_44978_CRC_CRC_TR_PW_	157.45 10	mg SO4/L	0.140363		x5.0000		UTESE	20
63	U53	439	7_65_439_44978_CRC_CRC_TR_PW_	240.88 20	mg SO4/L	0.068586		x20.0000		UTESE	20
63	U53	439		0.00	mg SO4/L	0.000000				UTESE	
64	U54	440	7_66_440_44978_CRC_CRC_TR_PW_	240.84 45	mg SO4/L	0.068577		x20.0000		UTESE	20

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Cup	Type	ID	Details	Result	Units	Raw data	QC Pro	Auto dil.	Man dil.	User	T

Tech Support

Test parameters Sulfate

Test number	2		
Long test name	Sulfate	**Short test name**	SO4
Units	mg SO4/L	**Decimal places**	2
Test type	End Point		

Main parameters

Sample volume	400	**Water volume**	0	**Number of mixes**	2
Cuvette primes	2	**Cuvette washes**	2	**Baseline on wash**	Yes
Cadmium volume	0	**Reduction time**	0	**Diluent location**	Reagent 24

Math parameters

Reaction time	270	**Wavelength**	405nm	**Polynomial order**	2
Repeat delta check	0.250	**Auto dilution factor**	1.000	**Manual dilution factor**	0.000
Linearity Low	0.0036	**Linearity High**	0.1603	**Latest Slope**	1.0000
Alert Low	0.000	**Alert High**	0.000	**Latest Intercept**	0.0000

Reagent parameters

Number of reagents 1

Reagent	Name	Volume	Delay	Replaced in blank
1	Turbidimetric reagent (sul	200	0	Yes

Standards

Auto Std. Number	10	**Auto Std. Concentration**	40.0000	**Exclude the Blank**	No
Auto Std. Position	Cup 5	**Standard 1 (Blank) Position**	Reagent 24	**Correlation limit**	0.9980

Standard	Value	Standard	Value	Auto Standard	Percentage	Value
1	0.0000			S90	1.0000	0.4000
				S91	3.0000	1.2000
				S92	5.0000	2.0000
				S93	15.0000	6.0000
				S94	25.0000	10.0000
				S95	40.0000	16.0000
				S96	50.0000	20.0000
				S97	75.0000	30.0000
				S98	85.0000	34.0000
				S99	100.0000	40.0000

QC PRO

Control	Low value	High value	Location	Frequency	Start before
CCV	16.0000	24.0000	Reagent 11	10	1
CCB	-0.8000	0.8000	Reagent 24	10	1

Control	%% (+ \ -)	Stock Conc.	Spike added	Location
DUP	0.0000			
SPKA	0.0000	0.0000	0.0000	Cup 105
SPKM	0.0000		0.0000	
SPKD	0.0000			

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Advanced parameters

Set diagnostic slope & intercept	No	**Extra debubble action**	Yes
Eliminate air from transfer	No	**Eliminate air from cuvette wash**	No
Extra aspiration steps	0	**Smart dilution**	No

Diagnostics

Cup	Type	Result	Absorbance	T	W
1		17.59	0.091964	0.108234	0.016270
2		0.57	0.003307	0.018842	0.015535
3		41.47	0.050608	0.066256	0.015648
15	U5	64.82	0.143215	0.158747	0.015532
16	U6	52.58	0.061521	0.076065	0.014544
19	U9	46.65	0.055793	0.070468	0.014675
22	U12	57.98	0.066544	0.082238	0.015694
28	U18	51.52	0.121746	0.137486	0.015740
43	U33	34.51	0.090640	0.106349	0.015710
44	U34	46.86	0.113700	0.129485	0.015785
46	U36	42.88	0.106559	0.122143	0.015583
48	U38	248.39	0.070275	0.084852	0.014577
49	U39	17.92	0.093242	0.108826	0.015584
50	U40	0.60	0.003544	0.018997	0.015453
51	U41	381.85	0.097789	0.112431	0.014641
52	U42	408.48	0.102812	0.117446	0.014634
53	U43	251.40	0.070948	0.085606	0.014657
54	U44	323.20	0.086226	0.100905	0.014679
55	U45	366.99	0.094929	0.109576	0.014647
56	U46	294.68	0.080322	0.095028	0.014706
57	U47	307.37	0.082973	0.097556	0.014583
58	U48	321.97	0.085975	0.100637	0.014661
59	U49	171.04	0.148686	0.164826	0.016140
60	U50	200.09	0.059077	0.073749	0.014672
61	U51	17.79	0.092734	0.109089	0.016355
62	U52	0.56	0.003241	0.019534	0.016293
63	U53	236.11	0.067503	0.082194	0.014690
64	U54	157.45	0.140363	0.156932	0.016569
65	U55	240.88	0.068586	0.083294	0.014709
66	U56	240.84	0.068577	0.083238	0.014661
67	U57	187.39	0.096432	0.111137	0.014705
68	U58	178.25	0.053728	0.068389	0.014661
69x10.000	U59	217.64	0.063240	0.077926	0.014686
75	U65	79.47	0.085139	0.102070	0.016931
78	U68	71.25	0.078268	0.092920	0.014652
79	U69	76.39	0.082599	0.097224	0.014626
80	U70	15.64	0.084097	0.100769	0.016672
81	U71	0.64	0.003809	0.020539	0.016730
82	U72	157.94	0.140667	0.157349	0.016682
83	U73	179.79	0.093486	0.108158	0.014672
84	U74	164.87	0.087552	0.102234	0.014682
85x5.000	U75	170.51	0.089817	0.104532	0.014716
86	U76	241.68	0.116280	0.130932	0.014652
87	U77	280.32	0.077269	0.092016	0.014747
88	U78	51.94	0.122462	0.139460	0.016998

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Cup	Type	Result	Absorbance	T	W

Tech Support

Nitrate (NOx)

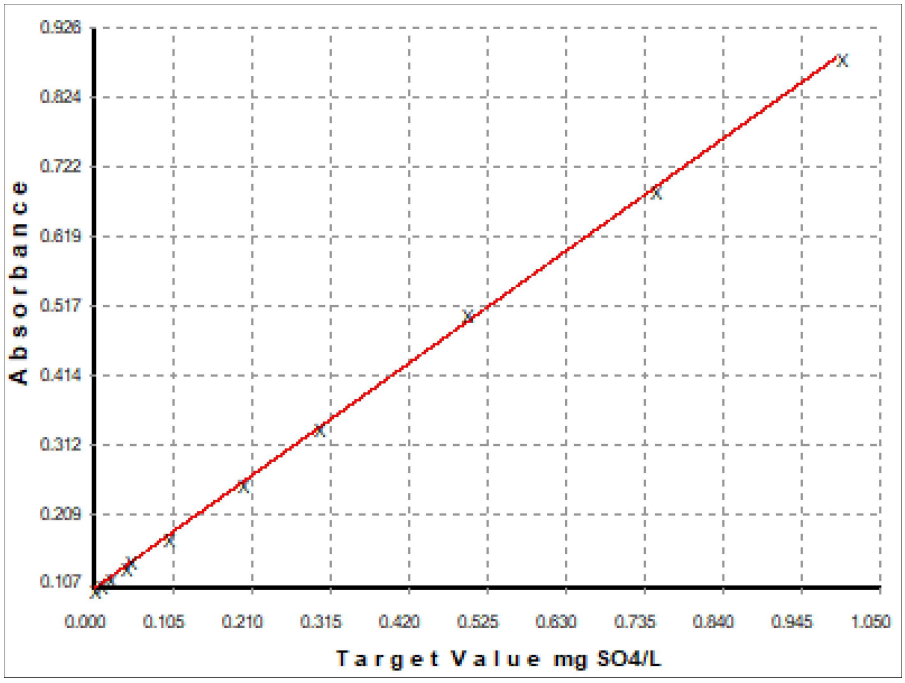
Calibration data

ID	Absorbance	Calc mg N/L	Target mg N/L	% Error
S1	0.1070	0.0002	0.0000	
S90	0.1127	0.0075	0.0111	-32.4831
S91	0.1241	0.0221	0.0222	-0.4682
S92	0.1387	0.0409	0.0444	-8.0292
S93	0.1495	0.0547	0.0500	9.3191
S94	0.1807	0.0946	0.1000	-5.3572
S95	0.2607	0.1970	0.2000	-1.5050
S96	0.3424	0.3016	0.3000	0.5337
S97	0.5113	0.5178	0.5000	3.5615
S98	0.6915	0.7486	0.7500	-0.1874
S99	0.8823	0.9928	1.0000	-0.7224
S0	0.1100	0.0041	0.0000	

Calibration status

Polynomial order	1
Correlation coefficient	0.9998
RSE (%)	12.51
Carryover(%)	0.4
Calibration equation	y = bx + a
y =	Concentration mg N/L
x =	Measured absorbance
a =	-1.367413E-01
b =	1.280267E+00
Original Date & Time	2025-03-10 14:17:39

Calibration graph



Reagents

Number	Reagent name	Batch number	User	Expiry date
13	DI Water 2			
5	Working color reagent (nitrate)			
39	DI Water (Blank)		UTESE	

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Test results

Cup	Type	ID	Details	Result	Units	Raw data	QC Pro	Auto dil.	Man dil.	U
	CCV	CCV		0.4445	mg N/L	0.454002				U
	CCB	CCB		-0.0110	mg N/L	0.098193				U
35	U25	411	7_32_411_44967_OWC_OWC_TR_RZ	2.7621	mg N/L	0.538293		x5.0000		U
36	U26	412	7_33_412_44967_OWC_OWC_TR_RZ	1.8447	mg N/L	0.394975		x5.0000		U
39	U29	415	7_36_415_44967_OWC_OWC_TR_RZ	1.1226	mg N/L	0.545242		x2.0000		U
40	U30	416	7_37_416_44967_OWC_OWC_TR_RZ	5.5491	mg N/L	0.540239		x10.0000		U
41	U31	417	7_38_417_44967_OWC_OWC_TR_RZ	1.5307	mg N/L	0.704607		x2.0000		U
49	U39	425	7_49_425_44967_OWC_OWC__Strear	6.7869	mg N/L	0.636923		x10.0000		U
68	U58	444	7_71_444_44978_CRC_CRC_UP_PW_	14.4629 10	mg N/L	0.671644		x20.0000		U
68	U58	444		0.0000	mg N/L	0.000000				U
69	U59	445	7_72_445_44978_CRC_CRC_UP_PW_	18.7411 20	mg N/L	0.253191		x10.0000	x10.0000	U
70	U60	446 Sample 447 Empty	7_74_446_44978_CRC_CRC_UP_PW_	7.3103 10	mg N/L	0.163907		x10.0000	x10.0000	U
71	U61	448	7_78_448_44978_CRC_CRC__Open__	2.4197	mg N/L	0.484802		x5.0000		U
	CCV	CCV		0.4582	mg N/L	0.464692				U
	CCB	CCB		0.0010	mg N/L	0.107601				U
72	U62	449	7_79_449_44978_CRC_CRC__Open__	2.2818	mg N/L	0.463263		x5.0000		U
73	U63	450	7_80_450_44978_CRC_CRC__Open__	2.4790	mg N/L	0.494077		x5.0000		U
75	U65	452	7_81_452_44978_CRC_CRC_TR_RZ_	0.9695	mg N/L	0.485458		x2.0000		U
81	U71	458	7_91_458_44978_CRC_CRC_UP_RZ_	14.1780	mg N/L	0.660520		x20.0000		U
82	U72	459	7_92_459_44978_CRC_CRC_UP_RZ_	22.0465	mg N/L	0.489478		x45.0000		U
82	U72	459		0.0000	mg N/L	0.000000				U
83	U73	460	7_93_460_44978_CRC_CRC_UP_RZ_	27.3807	mg N/L	0.582066		x45.0000		U
83	U73	460		0.0000	mg N/L	0.000000				U
84	U74	461	7_94_461_44978_CRC_CRC_UP_RZ_	10.7282	mg N/L	0.525791		x20.0000		U
85	U75	462 Samples 463 and 464 Empt	7_95_462_44978_CRC_CRC_UP_RZ_	3.7674	mg N/L	0.224513		x5.0000	x5.0000	U
86	U76	465	7_98_465_44978_CRC_CRC_UP_RZ_	22.6795	mg N/L	0.500465		x45.0000		U
86	U76	465		0.0000	mg N/L	0.000000				U
87	U77	Field Blank		0.0970	mg N/L	0.182549				U
	CCV	CCV		0.4644	mg N/L	0.469509				U
	CCB	CCB		0.0082	mg N/L	0.113204				U
88	U78	Field Blank		0.0761	mg N/L	0.166238				U
89	U79	Field Blank		0.8881	mg N/L	0.800494				U
	CCV	CCV		0.4616	mg N/L	0.467363				U
	CCB	CCB		0.0147	mg N/L	0.118255				U

Test parameters Nitrate (NOx)

Test number	3		
Long test name	Nitrate (NOx)	**Short test name**	NOx_NO3
Units	mg N/L	**Decimal places**	4
Test type	End Point		

Main parameters

Sample volume	180	**Water volume**	0	**Number of mixes**	1
Cuvette primes	2	**Cuvette washes**	2	**Baseline on wash**	Yes
Cadmium volume	0	**Reduction time**	0	**Diluent location**	Reagent 24

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SEAL Analytical

Math parameters

Reaction time	1825	**Wavelength**	520nm	**Polynomial order**	1
Repeat delta check	0.250	**Auto dilution factor**	1.000	**Manual dilution factor**	0.000
Linearity Low	0.1070	**Linearity High**	0.8823	**Latest Slope**	1.2803
Alert Low	0.000	**Alert High**	0.000	**Latest Intercept**	-0.1367

Reagent parameters

Number of reagents 2

Reagent	Name	Volume	Delay	Replaced in blank
1	DI Water 2	10	0	No
2	Working color reagent (nit310)		27	No

Standards

Auto Std. Number	10	**Auto Std. Concentration**	1.0000	**Exclude the Blank**	No
Auto Std. Position	Cup 3	**Standard 1 (Blank) Position**	Reagent 24	**Correlation limit**	0.9980

Standard	Value	Standard	Value	Auto Standard	Percentage	Value
1	0.0000			S90	1.1111	0.0111
				S91	2.2222	0.0222
				S92	4.4444	0.0444
				S93	5.0000	0.0500
				S94	10.0000	0.1000
				S95	20.0000	0.2000
				S96	30.0000	0.3000
				S97	50.0000	0.5000
				S98	75.0000	0.7500
				S99	100.0000	1.0000

QC PRO

Control	Low value	High value	Location	Frequency	Start before
CCV	0.4000	0.6000	Reagent 12	10	1
CCB	-0.0200	0.0200	Reagent 24	10	1

Control	%% (+ \ -)	Stock Conc.	Spike added	Location
DUP	0.0000			
SPKA	0.0000	3.0000	0.2500	Cup 103
SPKM	0.0000		0.0000	
SPKD	0.0000			

Advanced parameters

Set diagnostic slope & intercept	No	**Extra debubble action**	No
Eliminate air from transfer	No	**Eliminate air from cuvette wash**	No
Extra aspiration steps	0	**Smart dilution**	No

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Diagnostics

Cup	Type	Result	Absorbance	T	W
1		0.4445	0.454002	0.458732	0.004729
2		-0.0110	0.098193	0.102775	0.004582
31	U21	2.7621	0.538293	0.543034	0.004741
32	U22	1.8447	0.394975	0.399760	0.004784
35	U25	1.1226	0.545242	0.550195	0.004953
36	U26	5.5491	0.540239	0.545533	0.005294
39	U29	1.5307	0.704607	0.710291	0.005683
47	U37	6.7869	0.636923	0.642670	0.005747
70x10.00(	U60	14.4629	0.671644	0.679872	0.008227
71	U61	18.7411	0.253191	0.259184	0.005993
72	U62	7.3103	0.163907	0.170021	0.006114
75	U65	2.4197	0.484802	0.491219	0.006417
76	U66	0.4582	0.464692	0.471292	0.006600
77	U67	0.0010	0.107601	0.114350	0.006749
78	U68	2.2818	0.463263	0.470109	0.006846
79	U69	2.4790	0.494077	0.501265	0.007188
83	U73	0.9695	0.485458	0.492758	0.007300
91		14.1780	0.660520	0.668063	0.007542
92		22.0465	0.489478	0.497604	0.008125
93		27.3807	0.582066	0.590348	0.008281
94		10.7282	0.525791	0.533772	0.007981
95		3.7674	0.224513	0.232587	0.008074
96		22.6795	0.500465	0.509058	0.008592
97		0.0970	0.182549	0.190989	0.008440
98		0.4644	0.469509	0.478350	0.008841
99		0.0082	0.113204	0.122104	0.008901
100		0.0761	0.166238	0.175251	0.009014
101		0.8881	0.800494	0.809811	0.009316
102		0.4616	0.467363	0.476935	0.009572
103		0.0147	0.118255	0.127824	0.009569

Tech Support

Phosphate

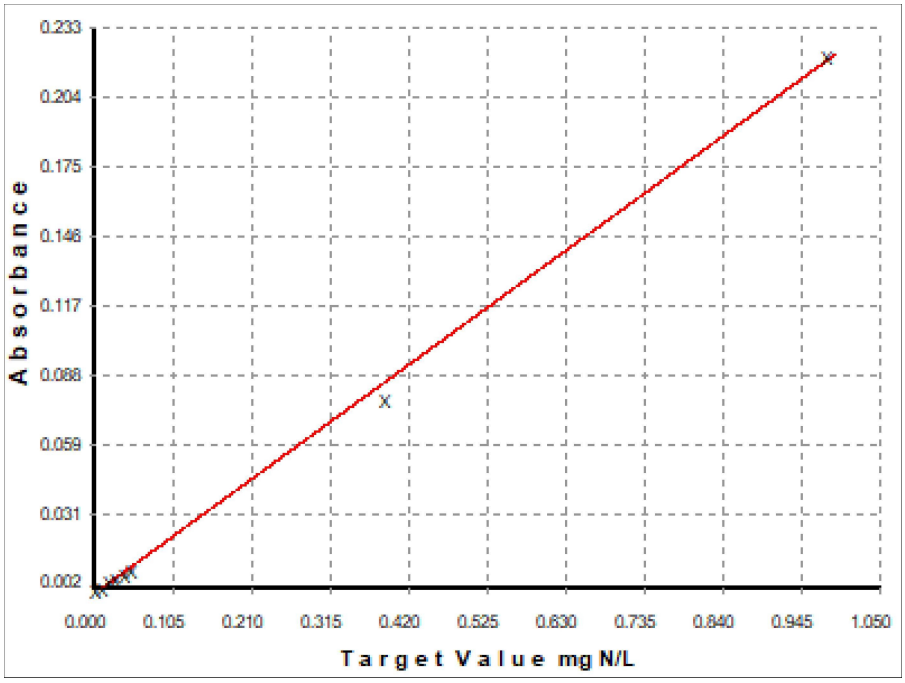
Calibration data

ID	Absorbance	Calc mg P/L	Target mg P/L	% Error
S1	0.0016	0.011	0.000	
S90	0.0026	0.015	0.010	50.0110
S91	0.0047	0.024	0.020	21.3811
S92	0.0061	0.031	0.030	1.9975
S93	0.0082	0.040	0.040	0.2920
S94	0.0098	0.047	0.050	-5.8946
S95	0.0801	0.360	0.390	-7.5759
*96	0.0947	0.426	0.400	6.4292
S97	0.2218	0.992	0.980	1.1987
*98	0.2439	1.090	0.990	10.1326
*99	0.2399	1.072	1.000	7.2439
S0	0.0023	0.014	0.000	

Calibration status

Polynomial order	1
Correlation coefficient	0.9993
RSE (%)	24.72
Carryover(%)	0.0
Calibration equation	y = bx + a
y =	Concentration mg P/L
x =	Measured absorbance
a =	3.555347E-03
b =	4.455637E+00
Original Date & Time	2025-03-10 19:07:44
	(2025-03-10 15:14:35)

Calibration graph



Reagents

Number	Reagent name	Batch number	User	Expiry date
6	Working color reagent (fosfate)			
7	Working ascorbic acid (fosfate)			
39	DI Water (Blank)		UTESE	

Tech Support

Test results

Cup	Type	ID	Details	Result	Units	Raw data	QC Pro	Auto dil.	Man dil.	U
	CCV	CCV		0.303	mg P/L	0.067127				U
	CCB	CCB		0.014	mg P/L	0.002292				U
11	U1	385	7_1_385_44967_OWC_OWC_W_PW_	0.012 10	mg P/L	0.001970				U
12	U2	386	7_2_386_44967_OWC_OWC_W_PW_	0.041 20	mg P/L	0.008359				U
13	U3	387	7_3_387_44967_OWC_OWC_W_PW_	0.049 45	mg P/L	0.010198				U
14	U4	388	7_4_388_44967_OWC_OWC_W_PW_	0.026 10	mg P/L	0.005063				U
15	U5	389	7_5_389_44967_OWC_OWC_W_PW_	0.023 20	mg P/L	0.004427				U
16	U6	390	7_6_390_44967_OWC_OWC_W_PW_	0.356 45	mg P/L	0.079023				U
17	U7	391	7_7_391_44967_OWC_OWC_W_PW_	0.352 10	mg P/L	0.078217				U
18	U8	392	7_8_392_44967_OWC_OWC_W_PW_	0.011 20	mg P/L	0.001654				U
19	U9	393	7_9_393_44967_OWC_OWC_W_PW_	0.114 45	mg P/L	0.024891				U
20	U10	394	7_11_394_44967_OWC_OWC_TR_PW	0.014 A_10	mg P/L	0.002359				U
	CCV	CCV		0.295	mg P/L	0.065500				U
	CCB	CCB		0.011	mg P/L	0.001635				U
21	U11	395	7_12_395_44967_OWC_OWC_TR_PW	0.013 A_20	mg P/L	0.002216				U
22	U12	396	7_13_396_44967_OWC_OWC_TR_PW	0.013 A_45	mg P/L	0.002067				U
23	U13	397	7_14_397_44967_OWC_OWC_TR_PW	0.026 B_10	mg P/L	0.005107				U
24	U14	398	7_15_398_44967_OWC_OWC_TR_PW	0.018 B_20	mg P/L	0.003320				U
25	U15	399	7_16_399_44967_OWC_OWC_TR_PW	0.011 ELL	mg P/L	0.001603				U
26	U16	400	7_17_400_44967_OWC_OWC_TR_PW	0.021 C_10	mg P/L	0.003913				U
27	U17	401	7_18_401_44967_OWC_OWC_TR_PW	0.012 C_20	mg P/L	0.001835				U
28	U18	402 Sample 403 Empty	7_19_402_44967_OWC_OWC_TR_PW	0.011 ELL	mg P/L	0.001598				U
29	U19	404	7_23_404_44967_OWC_OWC_UP_PW	0.016 A_45	mg P/L	0.002850				U
30	U20	405	7_24_405_44967_OWC_OWC_UP_PW	0.030 B_10	mg P/L	0.005979				U
	CCV	CCV		0.304	mg P/L	0.067339				U
	CCB	CCB		0.009	mg P/L	0.001180				U
31	U21	406	7_25_406_44967_OWC_OWC_UP_PW	0.044 ELL	mg P/L	0.001158		x5.0000		U
32	U22	407	7_27_407_44967_OWC_OWC_UP_PW	0.089 C_10	mg P/L	0.019239				U
33	U23	408 Sample 409 Empty	7_28_408_44967_OWC_OWC_UP_PW	0.014 C_20	mg P/L	0.002233				U
34	U24	410	7_31_410_44967_OWC_OWC_TR_RZ	0.046 _	mg P/L	0.009631				U
35	U25	411	7_32_411_44967_OWC_OWC_TR_RZ	0.020 _	mg P/L	0.003702				U
36	U26	412	7_33_412_44967_OWC_OWC_TR_RZ	0.403 _	mg P/L	0.089709				U
37	U27	413	7_34_413_44967_OWC_OWC_TR_RZ	0.224 _	mg P/L	0.049460				U
38	U28	414	7_35_414_44967_OWC_OWC_TR_RZ	0.018 _	mg P/L	0.003208				U
39	U29	415	7_36_415_44967_OWC_OWC_TR_RZ	0.024 _	mg P/L	0.004503				U
40	U30	416	7_37_416_44967_OWC_OWC_TR_RZ	0.018 _	mg P/L	0.003303				U
	CCV	CCV		0.306	mg P/L	0.067782				U
	CCB	CCB		0.011	mg P/L	0.001615				U
41	U31	417	7_38_417_44967_OWC_OWC_TR_RZ	0.057 _	mg P/L	0.012029				U
42	U32	418	7_41_418_44967_OWC_OWC_UP_RZ	0.107 _	mg P/L	0.023264				U
43	U33	419	7_42_419_44967_OWC_OWC_UP_RZ	0.122 _	mg P/L	0.026478				U
44	U34	420	7_43_420_44967_OWC_OWC_UP_RZ	0.071 _	mg P/L	0.015181				U
45	U35	421	7_44_421_44967_OWC_OWC_UP_RZ	0.069 _	mg P/L	0.014641				U
46	U36	422	7_45_422_44967_OWC_OWC_UP_RZ	0.268 _	mg P/L	0.059259				U
47	U37	423	7_46_423_44967_OWC_OWC_UP_RZ	0.035 _	mg P/L	0.007095				U
48	U38	424	7_48_424_44967_OWC_OWC_UP_RZ	0.070 _	mg P/L	0.014834				U
49	U39	425	7_49_425_44967_OWC_OWC__Strear	0.022	mg P/L	0.004247				U

Tech Support

Cup	Type	ID	Details	Result	Units	Raw data	QC Pro	Auto dil.	Man dil.	U
50	U40	426	7_51_426_44978_CRC_CRC_W_PW_	0.027 10	mg P/L	0.005347				U
	CCV	CCV		0.319	mg P/L	0.070858				U
	CCB	CCB		0.013	mg P/L	0.002020				U
51	U41	427	7_52_427_44978_CRC_CRC_W_PW_	0.018 20	mg P/L	0.003168				U
52	U42	428	7_53_428_44978_CRC_CRC_W_PW_	0.013 45	mg P/L	0.002173				U
53	U43	429	7_54_429_44978_CRC_CRC_W_PW_	0.022 10	mg P/L	0.004069				U
54	U44	430	7_55_430_44978_CRC_CRC_W_PW_	0.015 20	mg P/L	0.002513				U
55	U45	431	7_56_431_44978_CRC_CRC_W_PW_	0.015 45	mg P/L	0.002532				U
56	U46	432	7_57_432_44978_CRC_CRC_W_PW_	0.070 10	mg P/L	0.014963				U
57	U47	433	7_58_433_44978_CRC_CRC_W_PW_	0.026 20	mg P/L	0.005082				U
58	U48	434	7_59_434_44978_CRC_CRC_W_PW_	0.019 45	mg P/L	0.003386				U
59	U49	435	7_61_435_44978_CRC_CRC_TR_PW_	0.014 _10	mg P/L	0.002404				U
60	U50	436	7_62_436_44978_CRC_CRC_TR_PW_	0.011 _20	mg P/L	0.001697				U
	CCV	CCV		0.321	mg P/L	0.071281				U
	CCB	CCB		0.012	mg P/L	0.001899				U
61	U51	437	7_63_437_44978_CRC_CRC_TR_PW_	0.012 _45	mg P/L	0.001904				U
62	U52	438	7_64_438_44978_CRC_CRC_TR_PW_	0.013 _10	mg P/L	0.002099				U
63	U53	439	7_65_439_44978_CRC_CRC_TR_PW_	0.042 _20	mg P/L	0.008658				U
64	U54	440	7_66_440_44978_CRC_CRC_TR_PW_	0.012 _45	mg P/L	0.001832				U
65	U55	441	7_67_441_44978_CRC_CRC_TR_PW_	0.018 _10	mg P/L	0.003284				U
66	U56	442	7_68_442_44978_CRC_CRC_TR_PW_	0.013 _20	mg P/L	0.002204				U
67	U57	443	7_69_443_44978_CRC_CRC_TR_PW_	0.014 _45	mg P/L	0.002416				U
68	U58	444	7_71_444_44978_CRC_CRC_UP_PW_	0.079 _10	mg P/L	0.002754		x5.0000		U
69	U59	445	7_72_445_44978_CRC_CRC_UP_PW_	0.116 _20	mg P/L	0.001807			x10.0000	U
70	U60	446 Sample 447 Empty	7_74_446_44978_CRC_CRC_UP_PW_	0.120 _10	mg P/L	0.001904			x10.0000	U
	CCV	CCV		0.307	mg P/L	0.068121				U
	CCB	CCB		0.011	mg P/L	0.001782				U
71	U61	448	7_78_448_44978_CRC_CRC__Open__	0.017	mg P/L	0.002916				U
72	U62	449	7_79_449_44978_CRC_CRC__Open__	0.017	mg P/L	0.002922				U
73	U63	450	7_80_450_44978_CRC_CRC__Open__	0.018	mg P/L	0.003319				U
74	U64	451	7_80_451_44978_CRC_CRC_TR_RZ_	0.016	mg P/L	0.002726				U
75	U65	452	7_81_452_44978_CRC_CRC_TR_RZ_	0.034	mg P/L	0.006842				U
76	U66	453	7_83_453_44978_CRC_CRC_TR_RZ_	0.024	mg P/L	0.004672				U
77	U67	454	7_85_454_44978_CRC_CRC_TR_RZ_	0.016	mg P/L	0.002811				U
78	U68	455	7_86_455_44978_CRC_CRC_TR_RZ_	0.108	mg P/L	0.023411				U
79	U69	456	7_87_456_44978_CRC_CRC_TR_RZ_	0.062	mg P/L	0.013024				U
80	U70	457	7_88_457_44978_CRC_CRC_TR_RZ_	0.032	mg P/L	0.006477				U
	CCV	CCV		0.319	mg P/L	0.070689				U
	CCB	CCB		0.012	mg P/L	0.001974				U
81	U71	458	7_91_458_44978_CRC_CRC_UP_RZ_	0.107	mg P/L	0.023300				U
82	U72	459	7_92_459_44978_CRC_CRC_UP_RZ_	0.132	mg P/L	0.028928				U
83	U73	460	7_93_460_44978_CRC_CRC_UP_RZ_	0.519	mg P/L	0.115665				U
84	U74	461	7_94_461_44978_CRC_CRC_UP_RZ_	0.586	mg P/L	0.130609				U
85	U75	462 Samples 463 and 464 Empt	7_95_462_44978_CRC_CRC_UP_RZ_	0.153	mg P/L	0.006084			x5.0000	U
86	U76	465	7_98_465_44978_CRC_CRC_UP_RZ_	0.044	mg P/L	0.009041				U
87	U77	Field Blank		0.096	mg P/L	0.020702				U
88	U78	Field Blank		0.103	mg P/L	0.022239				U
89	U79	Field Blank		0.022	mg P/L	0.004157				U
	CCV	CCV		0.327	mg P/L	0.072640				U
	CCB	CCB		0.016	mg P/L	0.002850				U

Tech Support

Cup	Type	ID	Details	Result	Units	Raw data	QC Pro	Auto dil.	Man dil.	U

Test parameters Phosphate

Test number	4		
Long test name	Phosphate	**Short test name**	PO4
Units	mg P/L	**Decimal places**	3
Test type	End Point		

Main parameters

Sample volume	400	**Water volume**	0	**Number of mixes**	2
Cuvette primes	4	**Cuvette washes**	2	**Baseline on wash**	Yes
Cadmium volume	0	**Reduction time**	0	**Diluent location**	Reagent 24

Math parameters

Reaction time	480	**Wavelength**	880nm	**Polynomial order**	1
Repeat delta check	0.250	**Auto dilution factor**	1.000	**Manual dilution factor**	0.000
Linearity Low	0.0016	**Linearity High**	0.2218	**Latest Slope**	4.4556
Alert Low	0.000	**Alert High**	0.000	**Latest Intercept**	0.0036

Reagent parameters

Number of reagents 2

Reagent	Name	Volume	Delay	Replaced in blank
1	Working color reagent (fo	140	0	No
2	Working ascorbic acid (fo	45te)	27	No

Standards

Auto Std. Number	10	**Auto Std. Concentration**	1.0000	**Exclude the Blank**	No
Auto Std. Position	Cup 4	**Standard 1 (Blank) Position**	Reagent 24	**Correlation limit**	0.9980

Standard	Value	Standard	Value	Auto Standard	Percentage	Value
1	0.0000			S90	1.0000	0.0100
				S91	2.0000	0.0200
				S92	3.0000	0.0300
				S93	4.0000	0.0400
				S94	5.0000	0.0500
				S95	39.0000	0.3900
				S96	40.0000	0.4000
				S97	98.0000	0.9800
				S98	99.0000	0.9900
				S99	100.0000	1.0000

Tech Support

QC PRO

Control	Low value	High value	Location	Frequency	Start before
CCV	0.3000	0.5000	Reagent 13	10	1
CCB	-0.0200	0.0200	Reagent 24	10	1
Control	**%% (+ \ -)**	**Stock Conc.**	**Spike added**	**Location**	
DUP	0.0000				
SPKA	10.0000	5.0000	0.2500	Cup 104	
SPKM	0.0000		0.0000		
SPKD	0.0000				

Advanced parameters

Set diagnostic slope & intercept	No	**Extra debubble action**	No
Eliminate air from transfer	No	**Eliminate air from cuvette wash**	No
Extra aspiration steps	0	**Smart dilution**	No

Diagnostics

Cup	Type	Result	Absorbance	T	W
1		0.303	0.067127	0.071757	0.004630
2		0.014	0.002292	0.006556	0.004264
3		0.012	0.001970	0.006171	0.004200
4		0.041	0.008359	0.012650	0.004291
5		0.049	0.010198	0.014674	0.004476
6		0.026	0.005063	0.009271	0.004209
7		0.023	0.004427	0.008541	0.004113
8		0.356	0.079023	0.085589	0.006566
9		0.352	0.078217	0.085606	0.007389
10		0.011	0.001654	0.007036	0.005382
11	U1	0.114	0.024891	0.030868	0.005977
12	U2	0.014	0.002359	0.007446	0.005088
13	U3	0.295	0.065500	0.072310	0.006810
14	U4	0.011	0.001635	0.006927	0.005292
15	U5	0.013	0.002216	0.007098	0.004882
16	U6	0.013	0.002067	0.006564	0.004497
17	U7	0.026	0.005107	0.010006	0.004899
18	U8	0.018	0.003320	0.007854	0.004534
19	U9	0.011 ELL	0.001603	0.006037	0.004433
20	U10	0.021	0.003913	0.008256	0.004342
21	U11	0.012	0.001835	0.006049	0.004214
22	U12	0.011 ELL	0.001598	0.005496	0.003898
23	U13	0.016	0.002850	0.006730	0.003880
24	U14	0.030	0.005979	0.010204	0.004225
25	U15	0.304	0.067339	0.073771	0.006432
26	U16	0.009	0.001180	0.006189	0.005009
27	U17	0.044 ELL	0.001158	0.005786	0.004627
28	U18	0.089	0.019239	0.024116	0.004877
29	U19	0.014	0.002233	0.006538	0.004305
30	U20	0.046	0.009631	0.014027	0.004396
31	U21	0.020	0.003702	0.008352	0.004650

Tech Support

Cup	Type	Result	Absorbance	T	W
32	U22	0.403	0.089709	0.097117	0.007408
33	U23	0.224	0.049460	0.055865	0.006405
34	U24	0.018	0.003208	0.008573	0.005365
35	U25	0.024	0.004503	0.008880	0.004377
36	U26	0.018	0.003303	0.007814	0.004511
37	U27	0.306	0.067782	0.074236	0.006454
38	U28	0.011	0.001615	0.006589	0.004974
39	U29	0.057	0.012029	0.017191	0.005162
40	U30	0.107	0.023264	0.028322	0.005059
41	U31	0.122	0.026478	0.031297	0.004819
42	U32	0.071	0.015181	0.019745	0.004564
43	U33	0.069	0.014641	0.019166	0.004525
44	U34	0.268	0.059259	0.064526	0.005267
45	U35	0.035	0.007095	0.011466	0.004371
46	U36	0.070	0.014834	0.019092	0.004258
47	U37	0.022	0.004247	0.008132	0.003885
48	U38	0.027	0.005347	0.009243	0.003896
49	U39	0.319	0.070858	0.076723	0.005865
50	U40	0.013	0.002020	0.006362	0.004342
51	U41	0.018	0.003168	0.006951	0.003783
52	U42	0.013	0.002173	0.005951	0.003778
53	U43	0.022	0.004069	0.007772	0.003703
54	U44	0.015	0.002513	0.006094	0.003581
55	U45	0.015	0.002532	0.006038	0.003506
56	U46	0.070	0.014963	0.018896	0.003933
57	U47	0.026	0.005082	0.008587	0.003506
58	U48	0.019	0.003386	0.006915	0.003529
59	U49	0.014	0.002404	0.005749	0.003345
60	U50	0.011	0.001697	0.004974	0.003277
61	U51	0.321	0.071281	0.076619	0.005337
62	U52	0.012	0.001899	0.005889	0.003990
63	U53	0.012	0.001904	0.005521	0.003617
64	U54	0.013	0.002099	0.005583	0.003483
65	U55	0.042	0.008658	0.012366	0.003707
66	U56	0.012	0.001832	0.005176	0.003344
67	U57	0.018	0.003284	0.006631	0.003347
68	U58	0.013	0.002204	0.005480	0.003275
69x10.000	U59	0.014	0.002416	0.005441	0.003025
70x10.000	U60	0.079	0.002754	0.005949	0.003196
71	U61	0.116	0.001807	0.004881	0.003074
72	U62	0.120	0.001904	0.004951	0.003047
73	U63	0.307	0.068121	0.073424	0.005303
74	U64	0.011	0.001782	0.005736	0.003954
75	U65	0.017	0.002916	0.006416	0.003500
78	U68	0.017	0.002922	0.006312	0.003390
79	U69	0.018	0.003319	0.006730	0.003411
82	U72	0.016	0.002726	0.006010	0.003284
83	U73	0.034	0.006842	0.010190	0.003348
84	U74	0.024	0.004672	0.008099	0.003428
85x5.000	U75	0.016	0.002811	0.005877	0.003066
86	U76	0.108	0.023411	0.027410	0.003999
87	U77	0.062	0.013024	0.016822	0.003799

Tech Support

Cup	Type	Result	Absorbance	T	W
89	U79	0.319	0.070689	0.075934	0.005246
90		0.012	0.001974	0.006109	0.004135
91		0.107	0.023300	0.027568	0.004268
92		0.132	0.028928	0.033519	0.004591
93		0.519	0.115665	0.122341	0.006676
94		0.586	0.130609	0.137502	0.006893
95		0.153	0.006084	0.010947	0.004864
96		0.044	0.009041	0.013391	0.004350
97		0.096	0.020702	0.025174	0.004473
100		0.103	0.022239	0.026938	0.004699
101		0.022	0.004157	0.008156	0.003999
102		0.327	0.072640	0.078231	0.005591
103		0.016	0.002850	0.006584	0.003734